EXPERIMENT 40 · MICROSCALE PROCEDURE

Extraction and identification of fatty acids from corn oil

The same chemistry as the handout (saponify, acidify, extract, dry, make the methyl esters, identify them by TLC), done in two capped test tubes with about a twentieth of the oil. It takes about 50 minutes of bench time before the TLC and about 35 for the plate, uses sulfuric acid in place of perchloric acid, and stains with iodine at room temperature.

Work in pairs. Each pair runs one sample, with two screw-cap tubes and one TLC plate.

What is different from the printed handout. 0.1 mL of oil in a screw-cap tube instead of 2 mL in a flask (enough KOH to break every ester, about three times what is needed); 6 M HCl added dropwise instead of 2.5 mL concentrated HCl; extractions in the tube, with no separatory funnel; drying through a short column of Na_2SO_4 in a Pasteur pipet; 2% sulfuric acid in methanol instead of methanol : perchloric acid; the esters spotted straight from a small petroleum ether layer, with no second drying or evaporation; an iodine jar at room temperature instead of a 110 °C oven. The background, the TLC plate and its lanes, and the report sheet stay the same.

Caution. Goggles and gloves throughout. KOH in ethanol and 6 M HCl are corrosive; 2% H_2SO_4 in methanol is acidic and toxic. Petroleum ether and the TLC solvent are extremely flammable: no flames anywhere in the room, and all evaporation in the hood. Keep the iodine jar closed and open it only in the hood.

Before you start

1. Set a water bath (or heating block) to **60 °C** in the hood.
2. Label two clean, dry 16 × 125 mm screw-cap test tubes **A** and **B**, and a small vial or tube **C**.
3. **Make a drying column:** push a small plug of cotton into the narrow part of a Pasteur pipet with a wire or thin rod, add anhydrous Na_2SO_4 to a depth of about 1.5 cm, and clamp it upright over tube B. Wet it with 0.5 mL of petroleum ether and let that run through; pour it into the waste.
   The rinse settles the bed and washes out fine dust, so the first drops of extract come through clear.

Part A. Saponification and extraction of the fatty acids

4. Weigh tube A with its cap. Record the mass on the report sheet (1).
5. Add about **0.10 mL of corn oil** (about 90 mg) to the bottom of the tube with a graduated pipet or a Pasteur pipet, without touching the walls. Cap it and weigh it again (2). Work out the mass of oil (3).
6. Add **2.0 mL of 0.5 M KOH in ethanol**. Cap tightly, swirl, and keep the tube in the 60 °C bath for **15 minutes**, swirling every few minutes. The oil droplets disappear into one clear solution.
   Hydroxide breaks the three ester bonds of each triglyceride; ethanol dissolves both the oil and the KOH.
7. Let the tube cool to room temperature. Add **1 mL of water**. In the hood, add **6 M HCl dropwise**, swirling after each drop, until a drop of the mixture on pH paper reads 2 or below (about 0.2 to 0.3 mL; count your drops). The mixture turns cloudy.
   The acid puts the proton back on each carboxylate, giving neutral fatty acids, which do not dissolve in the watery mixture.

8. Add **2 mL of petroleum ether**. Cap and shake for 30 seconds, loosening the cap briefly after the first few shakes to let the pressure out. Stand the tube in the rack until two layers form.
9. With a Pasteur pipet, take off the **upper** (petroleum ether) layer, leaving the lower layer behind, and run it through the drying column into tube B.
10. Add another **2 mL of petroleum ether** to tube A, shake, let it separate, and run its upper layer through the same column into tube B. Rinse the column with **0.5 mL** of petroleum ether into tube B. Keep tube A until the end of the lab, then empty it into the aqueous waste.

    A second extraction recovers most of the fatty acid the first one left behind; the rinse carries what is still wetting the salt.

Part B. Preparation of the methyl esters

11. In the hood, stand tube B in a warm water bath (40 to 50 °C) and blow a gentle stream of air or nitrogen onto the surface of the liquid through a Pasteur pipet connected to the line, until only an oily film is left (about 5 minutes). Stop as soon as it is dry.
12. Add **2.0 mL of 2% sulfuric acid in methanol**. Cap tightly and keep tube B in the 60 °C bath for **15 minutes**.

    Fischer esterification: methanol, in large excess, turns each fatty acid into its methyl ester, catalyzed by the sulfuric acid.
13. Let the tube cool. Add **2 mL of water** and **0.5 mL of petroleum ether**, cap, shake, and let the layers separate.
14. Transfer the **upper** layer, which holds the methyl esters, to vial C and cap it. This is your sample for TLC.

    The water pulls the methanol and the acid into the lower layer. A trace of water does not matter on a TLC plate, so this layer is spotted without further drying.

Part C. Identification of the fatty acids by TLC

15. Pour about 15 mL of hexane : diethyl ether (4:1) into a 500 mL beaker, cover it with aluminum foil and let the air inside fill with vapor while you spot.
16. Handle the 15 × 6.5 cm silica plate only by its edges, with gloves. With a pencil, draw a light line about 1 cm from the edge, parallel to the 6.5 cm side, and mark five equally spaced spots (handout, Fig. 40.2).
17. With a separate capillary for each, spot: **lanes 1 and 5**, your methyl esters from vial C (touch the plate two or three times on the same spot, letting it dry in between); **lane 2**, methyl oleate; **lane 3**, methyl linoleate; **lane 4**, methyl palmitate. Keep each spot about 1 mm across. Let the spots dry.
18. Stand the plate in the beaker with the spots **above** the solvent, cover with the foil and leave it undisturbed.
19. When the solvent front is 1 to 2 cm from the top, take the plate out and **mark the front with a pencil at once**. Let the solvent evaporate in the hood for a minute or two.
20. Put the plate in the **iodine jar** (a jar with a few iodine crystals, closed since the start of the lab and kept at room temperature), close it, and wait **5 to 10 minutes**, until yellow-brown spots appear.
21. In the hood, take the plate out and **circle every spot in pencil immediately**: the color fades as the iodine evaporates. Close the jar again.
22. Measure the distance from the origin line to the solvent front (4) and to the center of each spot (5 to 9), and calculate the R_f values (10 to 14).

Waste

Lower layers from tubes A and B: aqueous waste (they hold ethanol or methanol, salts and a little acid). Petroleum ether and TLC solvent: flammable organic waste. Used Na_2SO_4 and cotton: solid waste. Capillaries and Pasteur pipets: broken glass box.

Chemicals and equipment

- Corn oil
- 0.5 M KOH in ethanol
- 6 M HCl, in a dropper bottle
- Petroleum ether (b.p. 30–60 °C)
- Anhydrous Na_2SO_4
- 2% (v/v) H_2SO_4 in methanol
- Methyl palmitate, methyl oleate, methyl linoleate (dilute solutions in petroleum ether)
- Hexane : diethyl ether (4:1)
- Iodine crystals, in a closed jar
- pH paper
- Two 16 × 125 mm screw-cap test tubes and a small capped vial
- 1 mL and 2 mL graduated pipets (or a pipettor)
- Pasteur pipets and bulbs; cotton; a thin rod
- Clamp and ring stand for the drying column
- Water bath or heating block at 60 °C
- Air or nitrogen line in the hood
- 15 × 6.5 cm silica gel TLC plate; capillary tubes
- 500 mL beaker and aluminum foil
- Ruler, pencil, gloves, analytical balance

Timing

Stage	Bench time
Set up, weigh the oil, saponify (15 min in the bath: make the drying column meanwhile)	about 20 min
Acidify, two extractions through the column	about 10 min
Evaporate, esterify (15 min in the bath: draw and mark the TLC plate meanwhile)	about 20 min
Extract the esters, spot, develop, stain, measure	about 35 min

Report sheet

Use the handout's report sheet as it is, reading "beaker" in lines 1 and 2 as "tube A with its cap."

Pre-lab questions

Questions 1, 2, 3 and 5 of the handout are unchanged. Question 4 is replaced:

4. The iodine jar is kept at room temperature and closed. Why must the spots be circled in pencil as soon as the plate comes out of the jar, and why is the jar opened only in the hood?